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Launch Speed and Spin versus Player Skill Level for Disc Golf Throws

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Introduction

Recently, the game of disc golf has seen rapid growth, with a 55% increase in Professional Disc Golf Association (PDGA) memberships in 2021 [1]. Recent papers have predicted the dynamics of golf discs, such as the model from Giljarhus *et al.* [2], but studies focusing on player performance have yet to be conducted. In 2017, Sasakawa *et al.* [3] compared ultimate forehand throws, a biomechanically similar action to forehand disc golf throws, from skilled and unskilled players. This abstract considers the performance characteristics of both backhand and forehand disc golf throws and relates them to player skill level.

Methods

Twenty-eight participants with varying skill levels took part in this study. Twenty-three of the participants were male and 5 were female. The official PDGA rating system was used to quantify the participants' skill levels [4]. The average male and female PDGA ratings from this study were 936 and 855, respectively, with higher ratings correlating to better-scoring players. The average participant for each sex was considered an above average amateur player [4]. The participants gave informed consent to take part in this study, and the study was conducted in accordance with the ethical standards of the University of Waterloo.

The players threw an Innova Disc Golf Destroyer (Innova Disc Golf, Rock Hill, USA) driver, with a mass of 173.5 grams, into a net. The players were allowed to take their preferred run-up to the throw. Five backhand and 5 forehand throws were recorded for each participant.

To quantify performance, a single high-speed Photron SA1 (Photron, Tokyo, Japan) camera, with a frame rate of 6250 frames/second and resolution of 1024 x 896 pixels, was mounted above the participants to record the disc motion at release. A minimum of 30 frames were used to estimate the speed and spin of the disc. Due to the single camera setup, the players were asked to throw the discs horizontally.

To find the disc's initial speed, it was assumed that the velocity of the disc was entirely within the horizontal plane and that the disc did not have any initial wobble. A circle was fitted around the disc in each frame to locate the disc centre and to estimate the pixel-to-metre conversion for a given throw. Regression lines relating the disc position (one for each of the frame axes) to time were calculated. The slopes of these lines were taken as the longitudinal and lateral velocity components to calculate the disc speed.

Under the same assumptions as above, the disc spin was estimated by tracking the rotation of circular markers located on the topside of the disc. The angles of lines from the markers to the disc centre were determined in each frame. Regression lines relating these angles to time were calculated and the average slope was taken as the disc spin.

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Results and Discussion

Table 1 shows the average launch speed and spin of the disc for varying skill levels. Within skill-level groupings, backhand throws produce significantly more spin but similar speed to forehand throws. In general, the higher-skilled players produced more speed and spin than the lower-skilled players. This agrees with the findings of Sasakawa *et al.* [3], where more skilful players generated higher speed and spin in forehand ultimate throws.

Table 1. Mean (relative standard deviation) of disc speed and spin grouped by different PDGA ratings.

	Number of	Backhand		Forehand	
PDGA Rating	**Participants**	**Speed (m/s)**	**Spin (rpm)**	**Speed (m/s)**	**Spin (rpm)**
<900	9	22.84 (11.9%)	920 (19.0%)	22.24 (15.1%)	656 (14.6%)
900-949	7	23.61 (7.8%)	1110 (9.7%)	24.06 (16.2%)	666 (14.2%)
950-999	10	26.87 (7.7%)	1252 (11.3%)	26.41 (5.4%)	734 (13.3%)
>999	2	30.16 (3.8%)	1332 (19.4%)	27.07 (11.8%)	774 (7.1%)
Total Range		19.62 – 30.97	713 – 1515	15.43 – 29.32	482 – 844

Table 2 shows the correlation coefficients of speed and spin with PDGA rating using the average backhand and forehand throws of each participant. Backhand speed and spin showed stronger correlations to PDGA rating than their forehand counterparts. Forehand spin for male participants was the weakest correlation and not significant. Most participants noted backhand as their preferred throwing style, meaning for some players, if the forehand is used sparingly during play, forehand throwing performance may not correlate well to their rating.

Table 2. Correlation coefficients, r, of speed and spin with PDGA rating. Unless denoted with *n.s.*, coefficients are significant ($P < 0.05$). ($r = 1$: perfect positive linear relationship; $r = 0$: no linear relationship exists).

Participants	Backhand Speed	Backhand Spin	Forehand Speed	Forehand Spin
Male only	0.61	0.64	0.57	0.36 ***n.s.***
All	0.72	0.77	0.63	0.43

To improve upon this work, more female participants should be included to produce statistically meaningful results for both sexes. Additionally, a stereo camera system can be used to collect the complete three-dimensional launch conditions of the disc. A trajectory model [2] can then be used to give further insight into the performance of the throw.

References

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